**Part 1:**

**Know our environment** (**earth-life supporting systems, chemistry of lithosphere, energy balance**, sustainability, and recycle)

know about global warming (infrared absorption, molecular vibration, atmospheric window, residence time of greenhouse gases, evidences and effects of global warming).

Future challenges ($CO_2$ sequestering and nuclear energy).

## Why is Earth Unique?

| Property | Earth | Mars | Venus |
|---|---|---|---|
| Water | ✓ | Limited | No |
| Oxygen | ✓ | Almost none | None |
| Average temperature | 15 °C | -63 °C | 460 °C |
| Life | Yes | No | No |

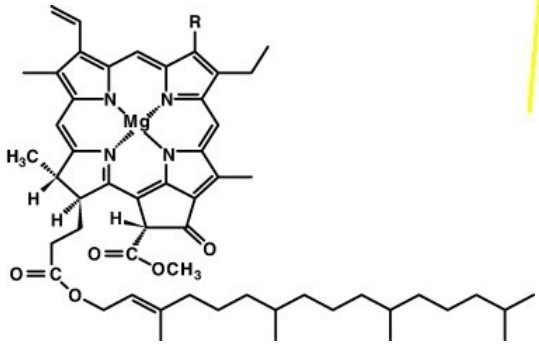
**If Mars had water, could humans survive there?**

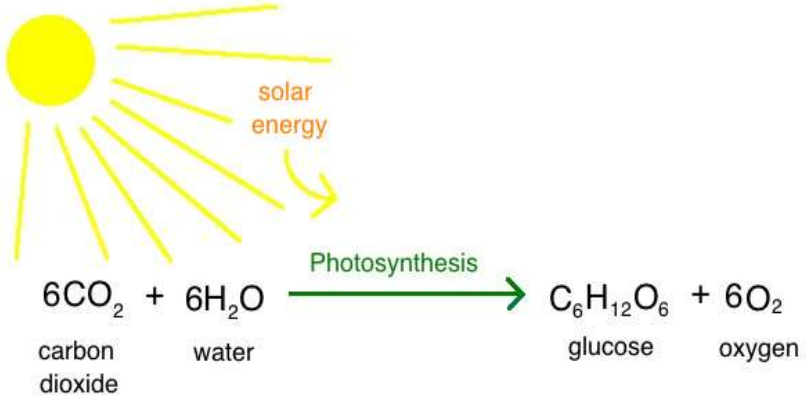
# What is Environment?

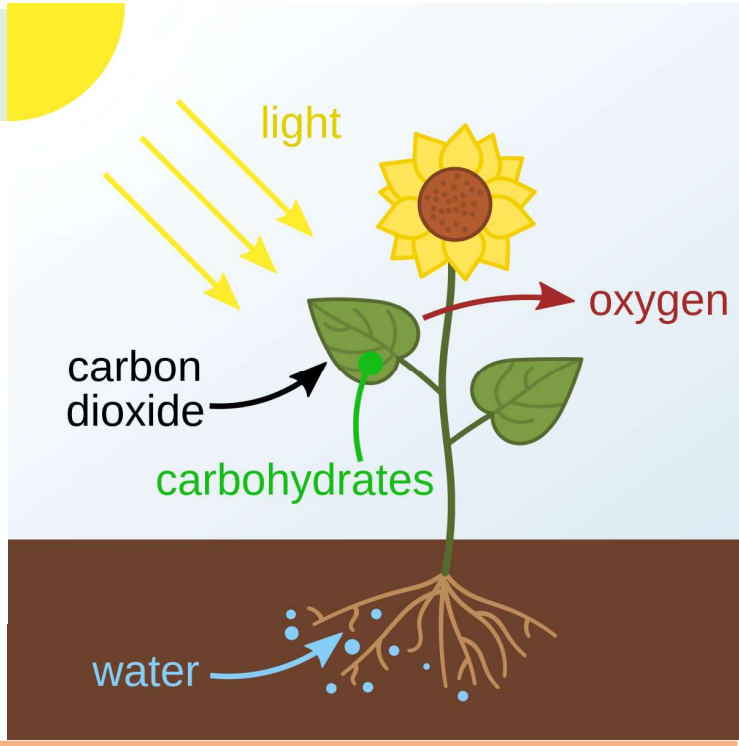
Many environmental problems and processes are chemical, and to understand them, basic chemistry involved is necessary

**Environment : Atmosphere-water-earth surroundings that make life possible**

Biological process of photosynthesis created the oxygen atmosphere in which we live in





# The Solar System

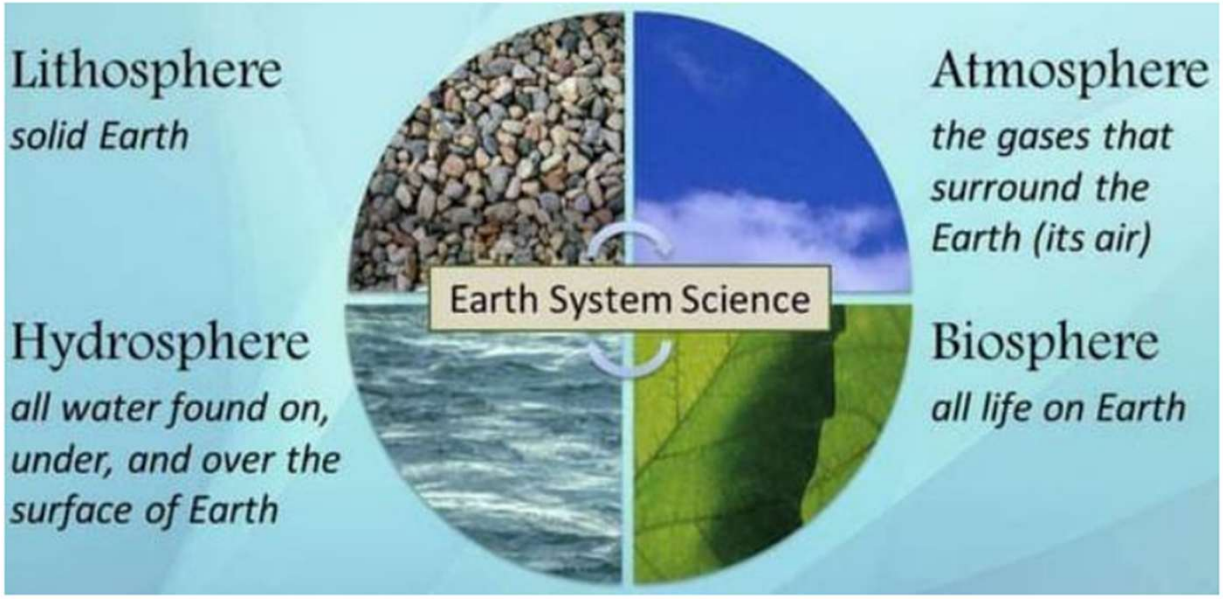
## Important Features of the Planets in Our Solar System

| Planet | Diameter (km) | Diameter (mi) | Mass (Earth = 1) | Density (water = 1) | Gravity (Earth = 1) | Time for One Rotation on Axis (Earth days or hours) | Time for One Revolution around Sun (Earth years) | Distance from Sun (million km) | Distance from Sun (million mi) | Composition of Atmosphere |
|---|---|---|---|---|---|---|---|---|---|---|
| **Terrestrial** | | | | | | | | | | |
| Mercury | 4,835 | 3,004 | 0.055 | 5.69 | 0.38 | 59 days | 0.24 | 57.7 | 36.8 | None |
| Venus | 12,194 | 7,577 | 0.815 | 5.16 | 0.89 | 243 days | 0.62 | 107 | 66.9 | $CO_2$ |
| Earth | 12,756 | 7,926 | 1.00 | 5.52 | 1.00 | 1.00 day | 1.0 | 149 | 92.6 | $N_2$, $O_2$ |
| Mars | 6,760 | 4,200 | 0.108 | 3.89 | 0.38 | 1.03 days | 1.9 | 226 | 141 | $CO_2$, $N_2$, Ar |
| **Giant** | | | | | | | | | | |
| Jupiter | 141,600 | 87,986 | 318 | 1.25 | 2.64 | 9.83 hours | 12 | 775 | 482 | $H_2$, He |
| Saturn | 120,800 | 75,061 | 95.1 | 0.62 | 1.17 | 10.23 hours | 29 | 1,421 | 883 | $H_2$, He |
| Uranus | 47,100 | 29,266 | 14.5 | 1.60 | 1.03 | 23.00 hours | 84 | 2,861 | 1,777 | $H_2$, He, $CH_4$ |
| Neptune | 44,600 | 27,713 | 17.0 | 2.21 | 1.50 | 22.00 hours | 165 | 4,485 | 2,787 | $H_2$, He, $CH_4$ |

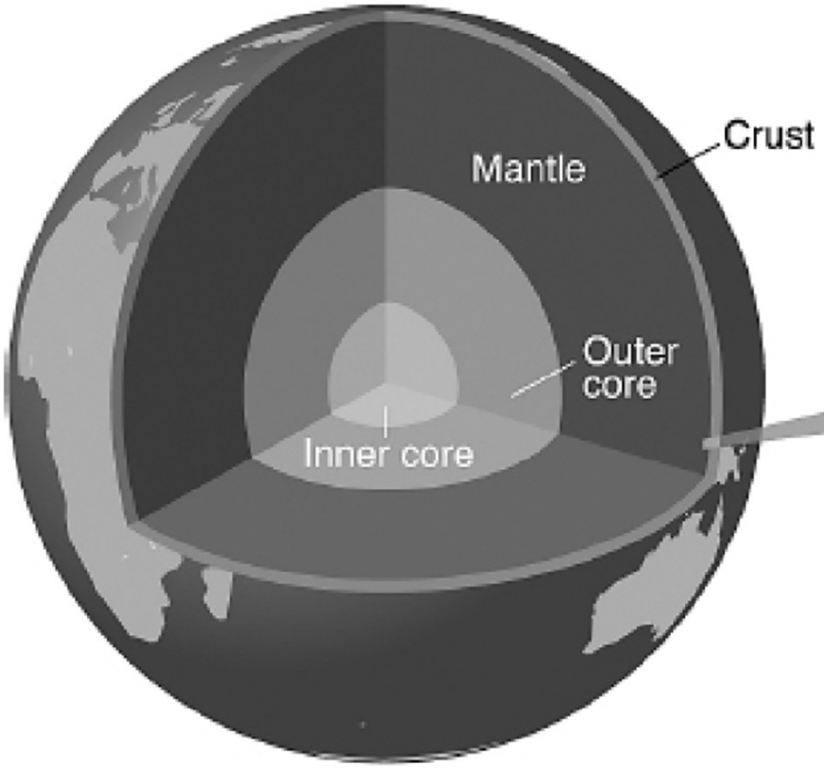
# Earth-life supporting systems - environment

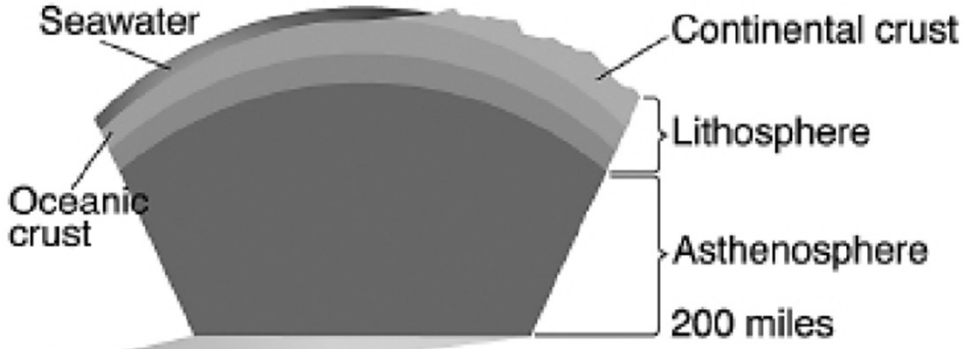
- Lithosphere (rock & soil)
- Atmosphere (air)
- Hydrosphere (water)
- Biosphere (living things)

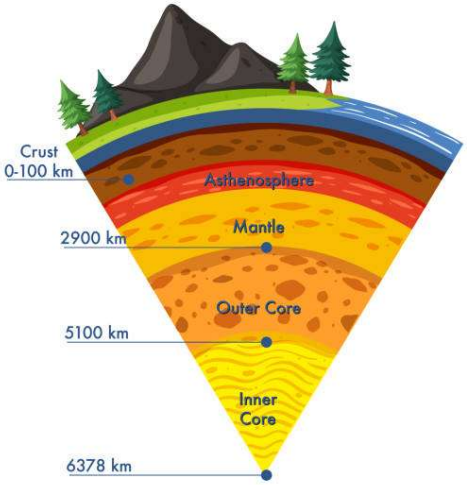
Life is sustained by the flow of energy from the sun through the biosphere, the cycling of nutrients within the biosphere and gravity.



# The structure of the Earth

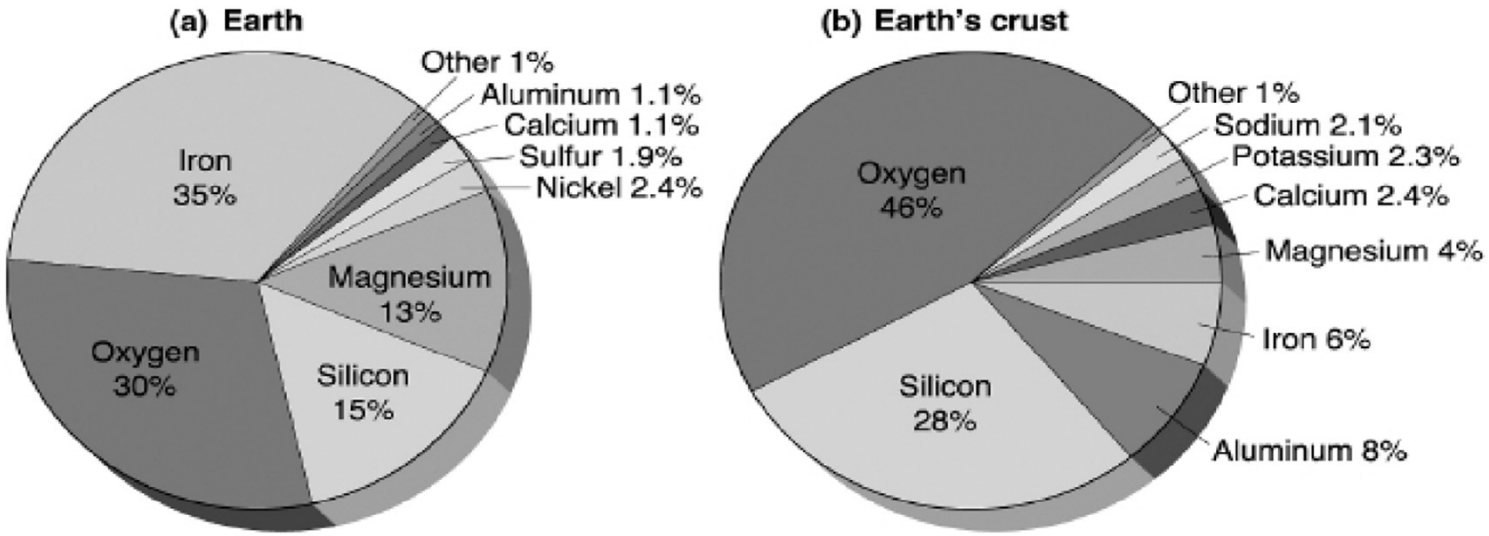


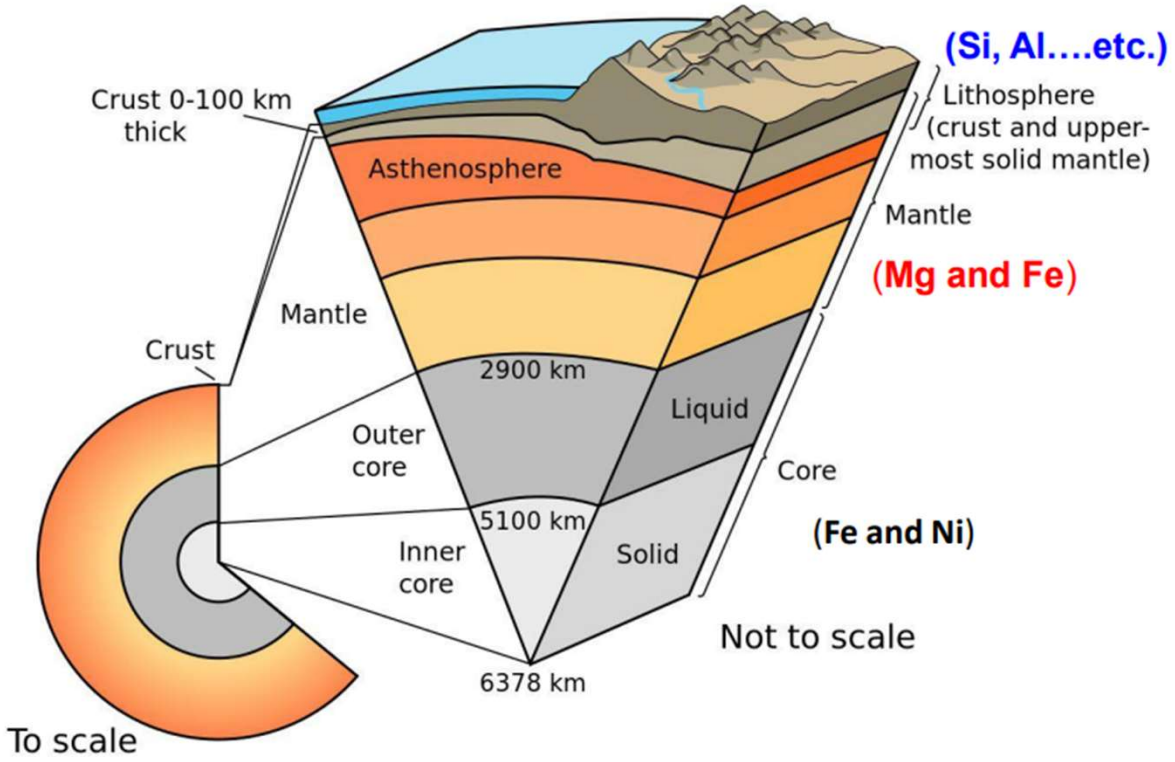

The lithosphere, which comprises the continental and oceanic crust together with the solid upper part of the mantle, rests on partially molten asthenosphere

# Lithosphere – Earth's crust




The relative abundance (by mass) of elements in the entire Earth and in the Earth's crust. Because of the differentiation that occurred early in the Earth's history, the percentage of iron in the crust (b) is lower than that in the whole Earth, (a) and the percentages of aluminum, silicon, and oxygen (the elements that combine to form silicates) are higher.

# Lithosphere – Earth's crust



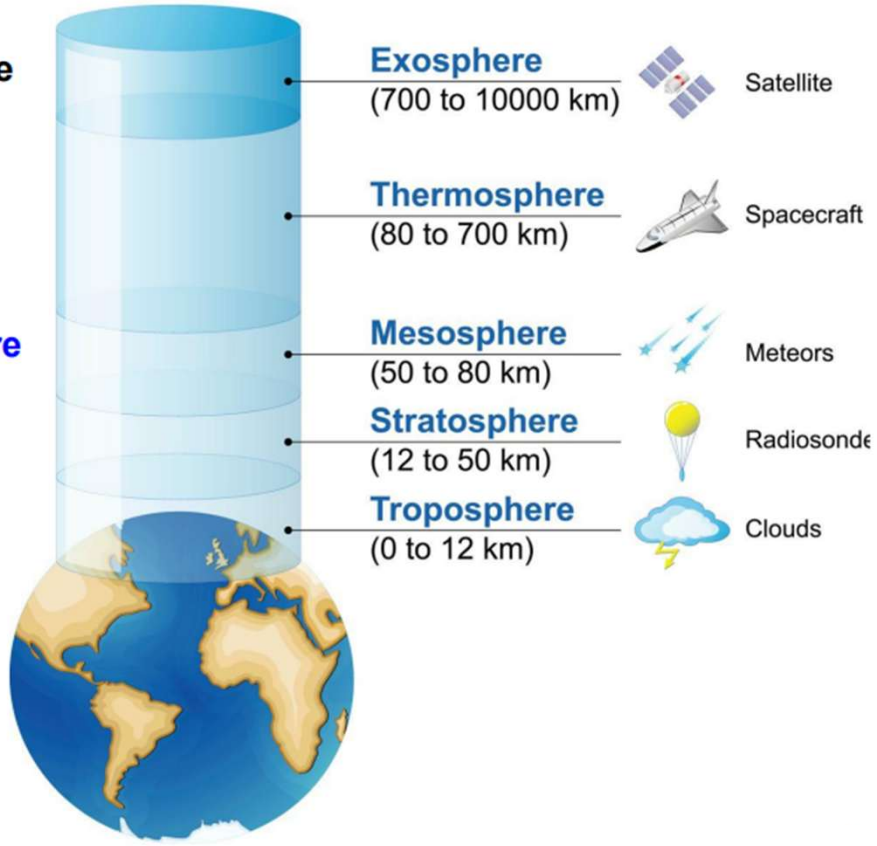

**The mass of the Earth ~ 6 x $10^{24}$ kg**

- Rocks
- Minerals
- Soil
- Fossil fuels
- Metal ores

**why soil formation takes thousands of years?**

# Atmosphere - Air

- **Exosphere**
  - Very high temperature
  - $H_2$, He
- **Thermosphere**
  - - 92 to 1200 °C
  - $O_2$, $NO^+$
- **Mesosphere or ionosphere**
  - - 2 to – 92 °C
  - $O_2^+$, $O^+$, $e^-$, $NO^+$
- **Stratosphere**
  - - 56 to - 2 °C
  - $O_3$ (ozone layer)
- **Troposphere**
  - 15 to – 56 °C
  - $N_2$, $O_2$, $CO_2$, $H_2O$




**Atmosphere**

•Provides oxygen for respiration.

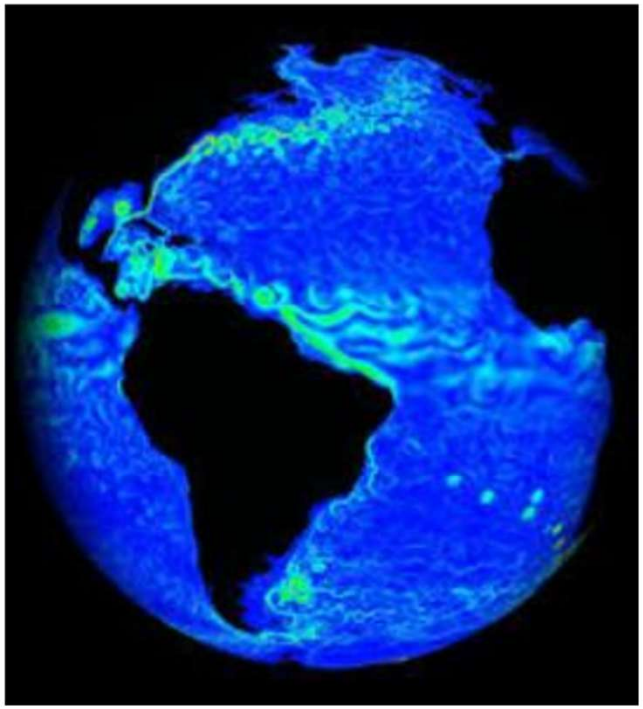
•Protects Earth from harmful UV radiation.

•Maintains Earth's temperature through the greenhouse effect.

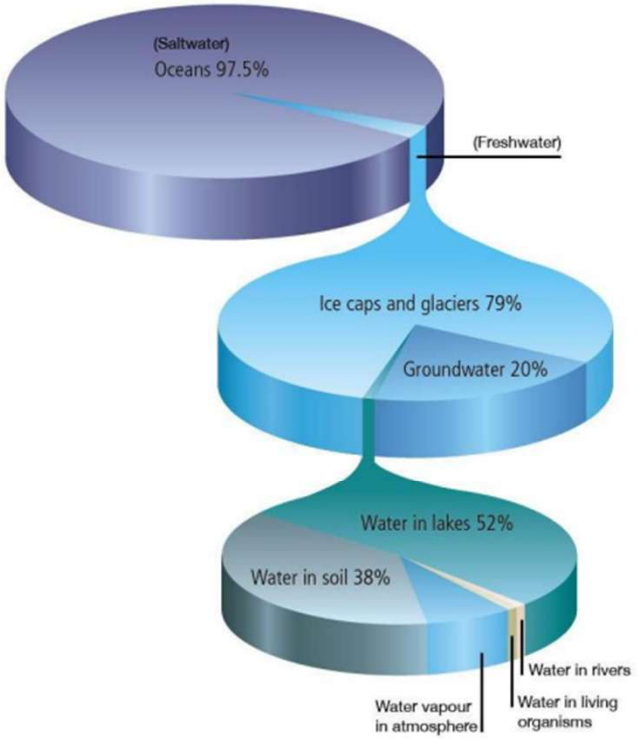
•Contains gases such as $N_2$ (78%), $O_2$ (21%), Ar (0.93%), and $CO_2$ (~0.04%).

**Without the atmosphere, Earth's average temperature would be about –18 °C instead of +15 °C.**

# Hydrosphere - Water

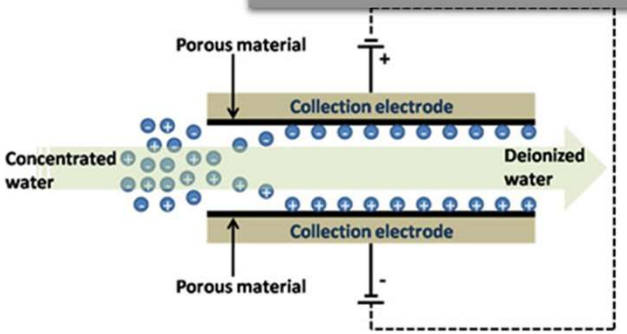


71% of Earth constitute of $H_2O$




Only 0.3% of it is usable by humans !

**Water purification by capacitive deionization**




Drinking water

- Oceans
- Rivers
- Lakes
- Groundwater
- Ice caps
- Atmospheric moisture
- Only **2.5%** of Earth's water is freshwater

# Reason for the salinity of ocean water

**Salinity is the total dissolved salt (TDS) in water.**

**Major ions that define salinity in ocean water**

| Ion | M (mol/L) | ppm |
|---|---|---|
| $Cl^-$ | $5.6 \times 10^{-1}$ | 19,000–19,500 |
| $Na^+$ | $4.8 \times 10^{-1}$ | 10,500–10,805 |
| $Mg^{2+}$ | $5.4 \times 10^{-2}$ | 1,290–1,350 |
| $SO_4^{2-}$ | $2.89 \times 10^{-2}$ | 2,688–2,700 |
| $Ca^{2+}$ | $1.05 \times 10^{-2}$ | 401–410 |
| $K^+$ | $1.05 \times 10^{-2}$ | 391 |
| $HCO_3^-$ | $2.4 \times 10^{-3}$ | 142–146 |
| $Br^-$ | $8.6 \times 10^{-4}$ | 67 |
| $BO_3^{3-}$ | $4.1 \times 10^{-4}$ | 24 |
| $Sr^{2+}$ | $9.51 \times 10^{-5}$ | 8 |
| $F^-$ | $7.5 \times 10^{-5}$ | 1.4 |

What is the % of $Na^+$ (weight/volume).

Parts per million and parts per billion

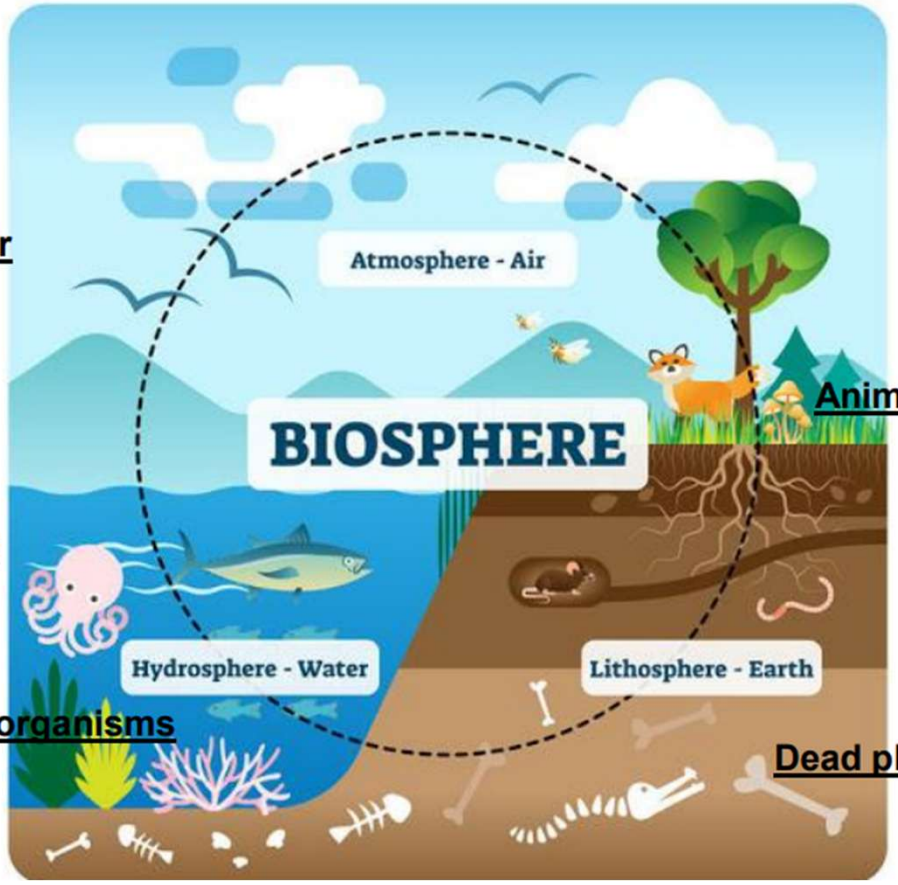
- Ocean water: 33-37 grams of dissolved salt per liter
- Drinking water: less than 300 mg per liter

# Classification of water based on salinity

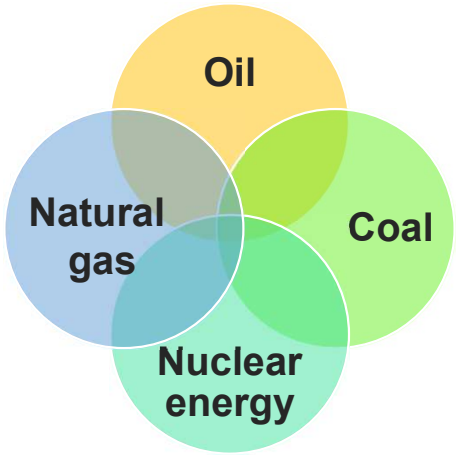
| Water type | Electrical conductivity, dS $m^{-1}$ | Salt concentration, mg $L^{-1}$ | Examples |
|---|---|---|---|
| Non-saline | <0.7 | <500 | Drinking, irrigation, and surface water |
| Slightly saline | 0.7–2 | 500–1,500 | Surface and irrigation water, and ground water |
| Moderately saline | 2–10 | 1,500–7,000 | Runoff and groundwater |
| Highly saline | 10–25 | 7,000–15,000 | Saline groundwater and runoff |
| Exceptionally saline | 25–45 | 15,000–35,000 | Very saline groundwater and seawater |
| Brine | >45 | >45,000 | Saline inland lakes and seas |



# Biosphere


Birds in the air
Atmosphere - Air
BIOSPHERE
Animals, Vegetation
Sea creatures
Hydrosphere - Water
Lithosphere - Earth
Marine living organisms
Dead plants & animals

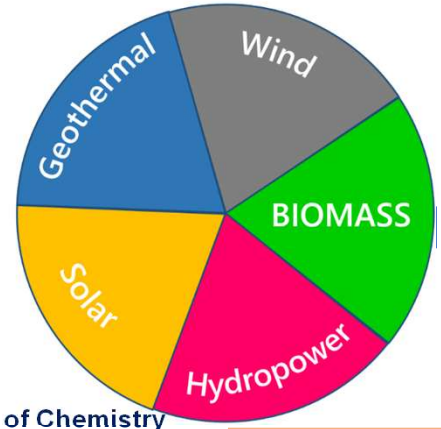
## Traditional Energy resources

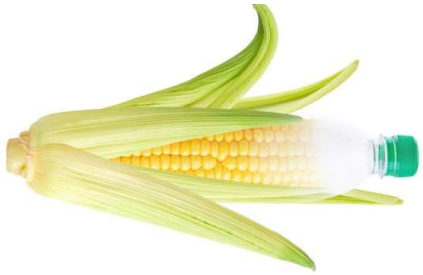



## Linear economy



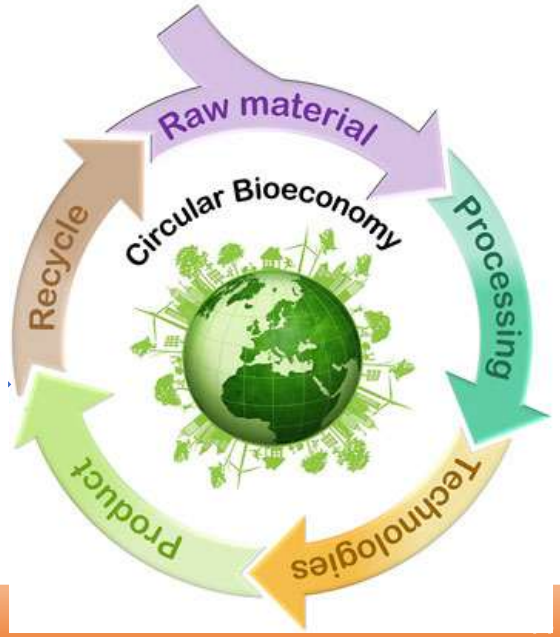

## Renewables



**Polylactic acid bioplastic**

Biomass (Carbon)

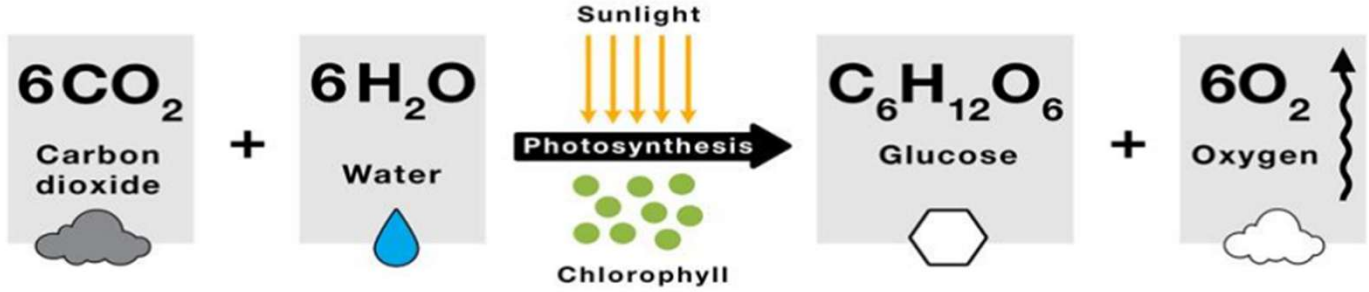


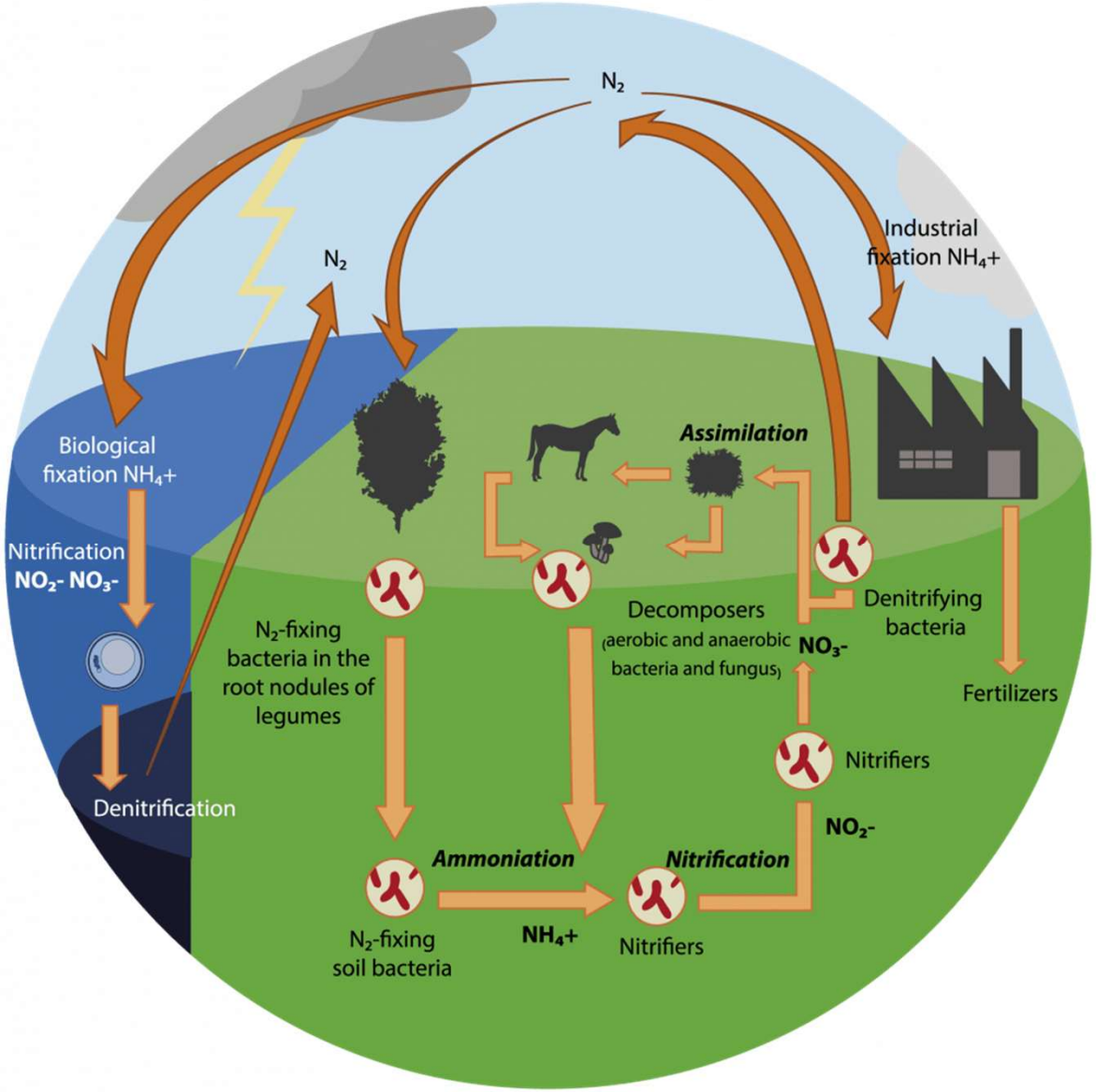
# Energy flow in the ecosystem

- **Photosynthesis: Solar energy is transformed into chemical energy** by photosynthesis. This energy is stored in plant tissue, and then transformed into mechanical and heat form during metabolic activities.

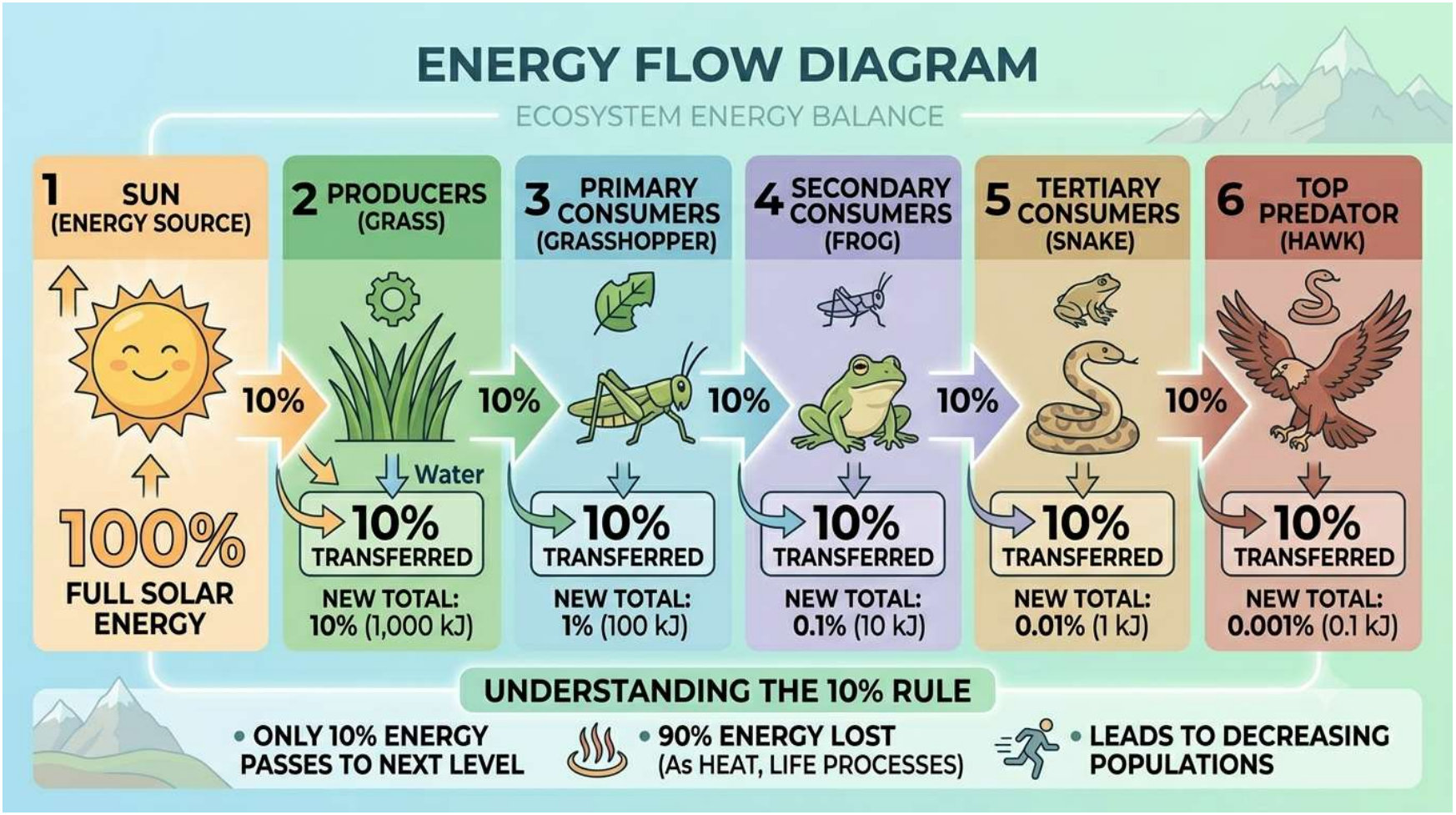
- In the biological world, the energy flows from sun to plants and then to all heterotrophic organisms like micro-organisms, animals, and human beings i.e. from producers to consumers. **1% of the total sunlight falling on the green plants is utilized in photosynthesis**.

- This is sufficient to maintain all life on this earth. There is no 100% flow of energy from producers to consumers. **Huge amount of energy** is always lost to environment in the form of another energy. Because of this, energy cannot be recycled in an ecosystem **'it can only flow one way'.**






One critical example is nitrogen, which makes up 78% of the atmosphere as nitrogen gas, but which is not usable in that form by most living things. Nitrogen cycles via the processes of nitrogen fixation, ammonification, nitrification, assimilation, decomposition, and denitrification.


ENERGY FLOW DIAGRAM
ECOSYSTEM ENERGY BALANCE
1 SUN (ENERGY SOURCE)
2 PRODUCERS (GRASS)
3 PRIMARY CONSUMERS (GRASSHOPPER)
4 SECONDARY CONSUMERS (FROG)
5 TERTIARY CONSUMERS (SNAKE)
6 TOP PREDATOR (HAWK)
10%
10%
10%
10%
10%
100%
FULL SOLAR ENERGY
Water
10% TRANSFERRED
NEW TOTAL: 10% (1,000 kJ)
10% TRANSFERRED
NEW TOTAL: 1% (100 kJ)
10% TRANSFERRED
NEW TOTAL: 0.1% (10 kJ)
10% TRANSFERRED
NEW TOTAL: 0.01% (1 kJ)
10% TRANSFERRED
NEW TOTAL: 0.001% (0.1 kJ)
UNDERSTANDING THE 10% RULE
• ONLY 10% ENERGY PASSES TO NEXT LEVEL
• 90% ENERGY LOST (As HEAT, LIFE PROCESSES)
• LEADS TO DECREASING POPULATIONS

# The energy flow follows the first and second laws of thermodynamics

## 1st law of thermodynamics:

The law states that energy can neither be created nor be destroyed but it can be transformed from one form to another. Similarly, solar energy utilized by green plants (producers) in photosynthesis converted into biochemical energy of plants and later into that of consumers.

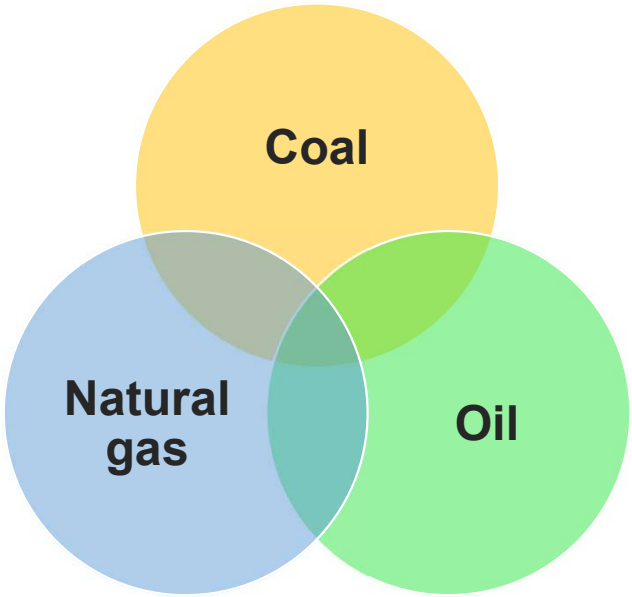
## 2nd law of thermodynamics:

The law states that the energy transformation involves degradation or dissipation of energy from a concentrated to a dispersed form. We have seen dissipation of energy occurs at every trophic level. There is loss of 90% energy, only 10% is transferred from one trophic level to the other.

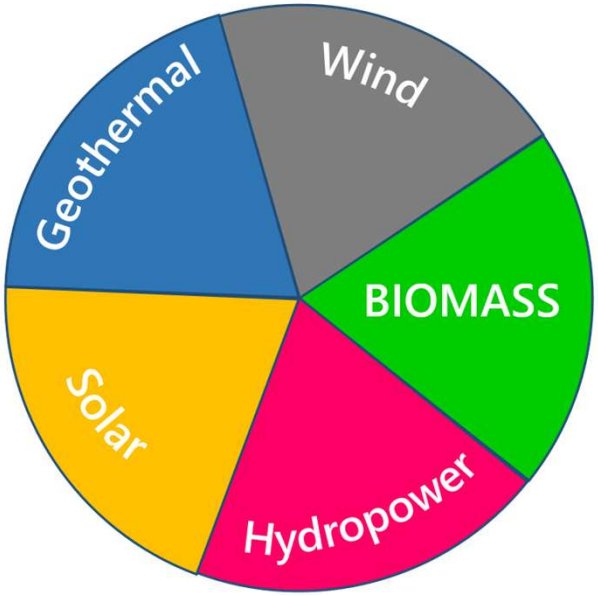
# Statements of second law: entropy

- **Energy loss in vehicles (chemical to mechanical):** Engine friction, wheels friction with roads, pumping air into and out of the engine, etc.
- The entropy of the Universe is increasing
  - $\Delta S_{Universe} > 0$
  - $\Delta S_{Universe} = \Delta S_{system} + \Delta S_{surroundings} > 0$
- In any process, if the entropy of the universe decreases, then the process will be an impossible process.

# Energy resources




Fossil fuels




Renewable energy

# How Fossil Fuels Are Formed

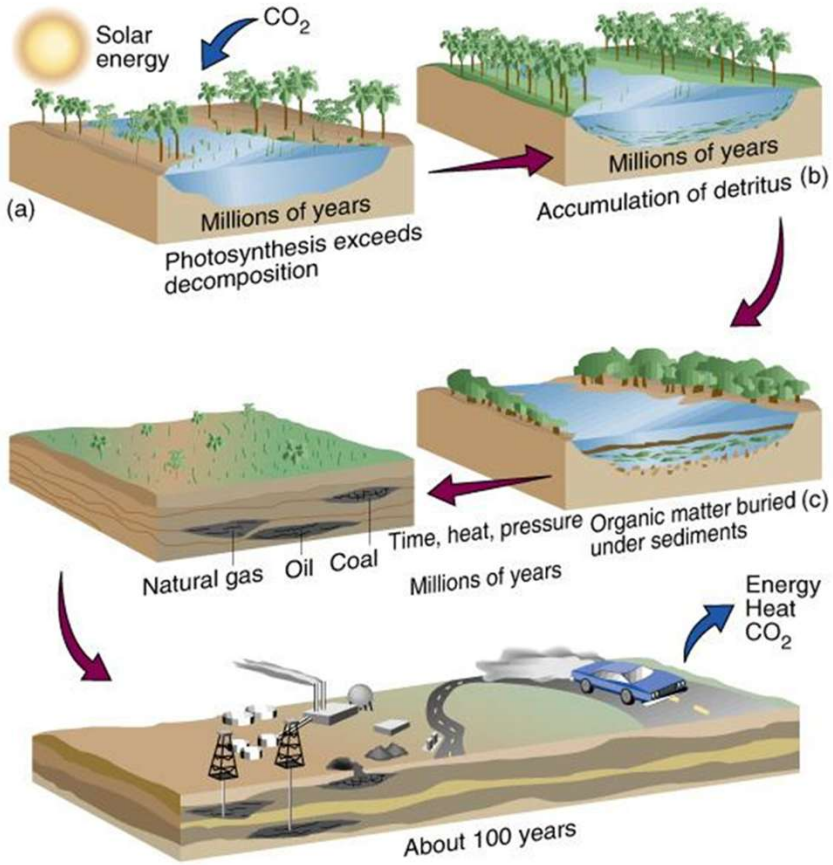
Millions of years of ….

1) photosynthesis exceeding respiration (decomposition).

2) detritus accumulation.

3) burial of detritus.

4) pressure & heat (metamorphosis).



# How is Energy derived from Fossil fuels?

By burning fossil fuels → Electricity + heat

$C_xH_y + (x+y/4)\ O_2 \rightarrow xCO_2 + y/2H_2O + heat$

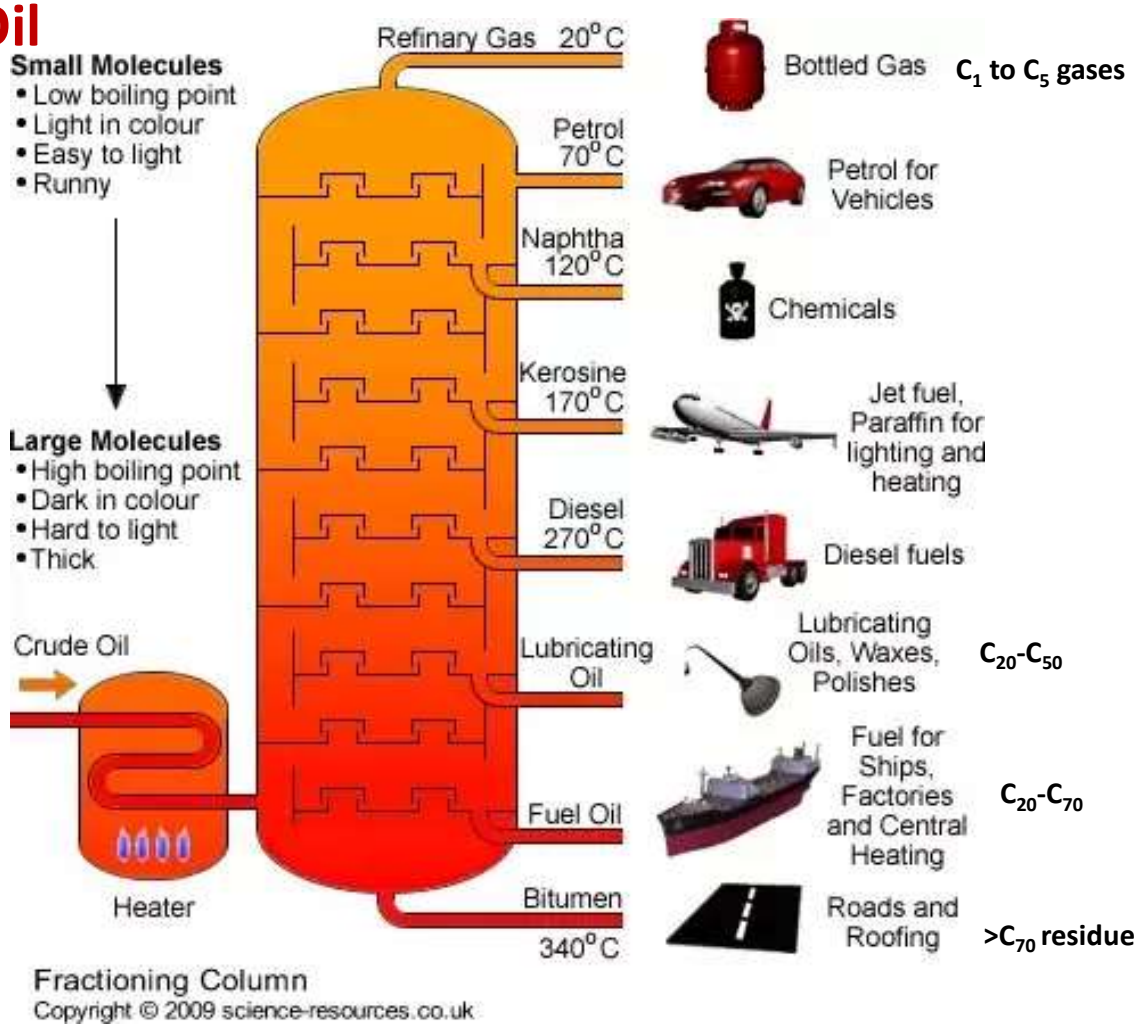
Electricity is generated by transforming mechanical energy (heat) to electrical energy in a turbine or generator

## Fossil fuel reserves

- Oil & Natural gas - 50 year supply
- Coal - 380-year supply
- Oil shales and oil sands - complex extraction technologies

# Fractional Distillation of Crude Oil

- Oil usually occurs with natural gas: mixture of hydrocarbons of light molecular weight



# Petroleum or crude oil

- Petroleum, or crude oil is a black, gooey liquid consisting of hundreds of different combustible hydrocarbons along with small amounts of sulfur, oxygen, and nitrogen impurities.
- Crude oil supplies about **one-third of the world's energy**.
- The products of **crude oil distillation,** called petrochemicals, are used as raw materials in industries.
- Conventional oil is currently abundant and relatively inexpensive and has a high net energy yield, but using it causes **air and water pollution** because they release **greenhouse gases**.





# Liquefied petroleum gas (LPG)

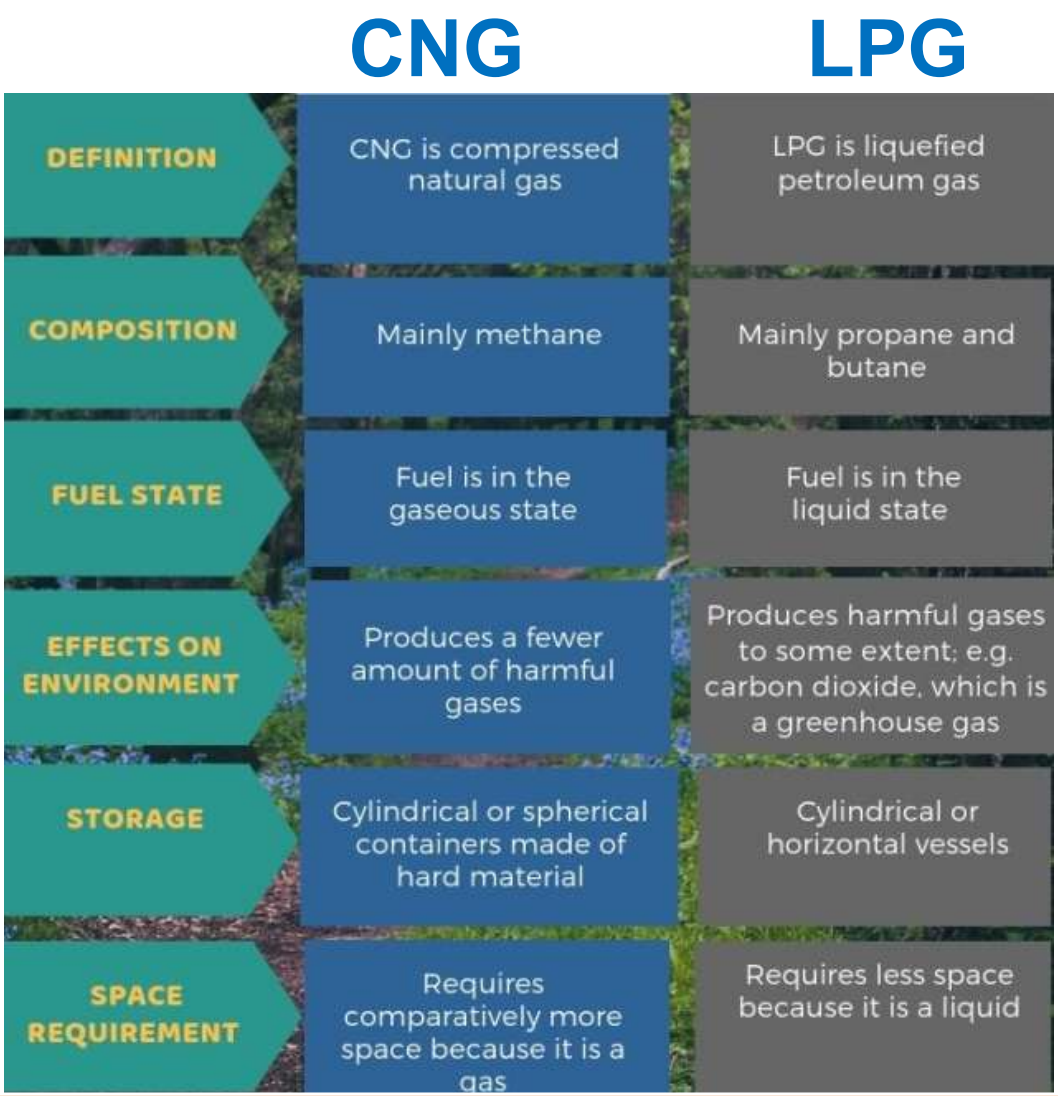


- The main components of LPG are **propane and butane,** and the other being lighter hydrocarbons.
- Under pressure (220-2200 kilopascals), petroleum is converted to LPG.
- It is a odorless gas.
- In domestic gas cylinders **mercaptan,** a foul smelling gas, is added so that leakage can be detected.

# Natural gas




- The main component of natural gas is **methane** in addition to various smaller amounts of other higher alkanes.
- It is a odorless gas.
- **CNG (compressed natural gas):** it is obtained at a pressure of 20–25 megapascals.
- A odorizer like **mercaptan**, a foul-smelling gas, is added so that leaks can be readily detected.
- It can be used in place of petrol (gasoline), diesel fuel, and liquefied petroleum gas (LPG).



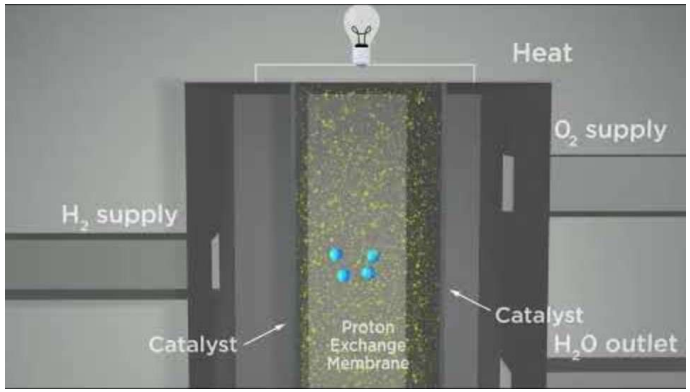


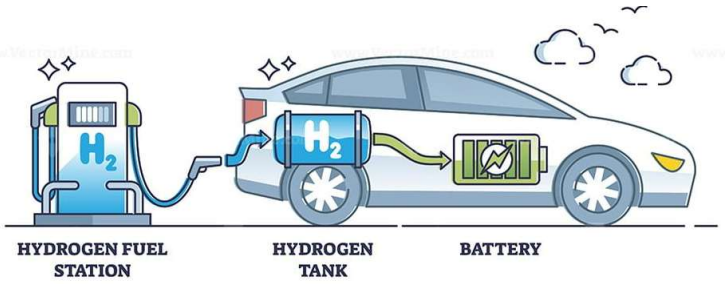

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भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

# Calorific Values of Fuels

| | |
|---|---|
| Hydrogen ($H_2$) | 120-142 MJ/kg |
| Methane ($CH_4$) | 50-55 MJ/kg |
| Methanol ($CH_3OH$) | 22.7 MJ/kg |
| Dimethyl ether - DME ($CH_3OCH_3$) | 29 MJ/kg |
| Petrol/gasoline | 44-46 MJ/kg |
| Diesel fuel | 42-46 MJ/kg |
| Crude oil | 42-47 MJ/kg |
| Liquefied petroleum gas (LPG) | 46-51 MJ/kg |
| Natural gas | 42-55 MJ/kg |
| Hard black coal (IEA definition) | >23.9 MJ/kg |
| Hard black coal (Australia & Canada) | c. 25 MJ/kg |
| Sub-bituminous coal (IEA definition) | 17.4-23.9 MJ/kg |
| Sub-bituminous coal (Australia & Canada) | c. 18 MJ/kg |
| Lignite/brown coal (IEA definition) | <17.4 MJ/kg |
| Lignite/brown coal (Australia, electricity) | c. 10 MJ/kg |
| Firewood (dry) | 16 MJ/kg |

# Hydrogen fuel cell electric vehicles




$H_2 + ½\ O_2 \longrightarrow$ Electrical energy + $H_2O$

**Anode:** Platinum catalyst enables hydrogen molecules to be split into protons and electrons.

**Cathode:** Platinum catalyst enables oxygen reduction by reacting with the protons generated by the anode, producing water.

**What is the catalyst and how is hydrogen produced?**

# Natural gas for hydrogen production



$H_2$ for fuel cell vehicles

- Natural gas (methane) is the main source of industrial hydrogen.
- Steam methane reforming (SMR) is used to producing hydrogen.
- SMR: $CH_4 + H_2O \xrightarrow{\Delta} CO + 3H_2$
- Nearly 50% of the world's hydrogen is produced by SMR method.
- A **fuel cell** can convert hydrogen into electrical energy for vehicles.
- Water is the exhaust emission in hydrogen vehicles.


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Indian Institute of Technology Hyderabad

# Coal



- Coal was formed 350 million years ago in the hot, damp regions of the earth (**lithosphere**).
- **Anthracite** [hard coal, 90% carbon, and the remaining is ash]
- **Bituminous** [soft coal, 80% carbon, and the remaining is ash]
- **Lignite** [brown coal, 70% carbon, and the remaining is ash]
- The coal reserves are likely to last for about 200 years, and if the usage increase by 2% per year, then it will last for another 65 years.
- India has **5% of world's coal,** but it is not very good in terms of heat capacity because of high ash content.

# Coal for hydrogen production




**$H_2$ for fuel cell vehicles**

- Coal gasification is used to produce hydrogen.
- **Coal gasification**

$$3C\ (\text{i.e., coal}) + O_2 + H_2O \xrightarrow{\Delta} H_2 + 3CO$$

- Nearly 20% of the world's hydrogen is produced by coal gasification method.





# Types of $H_2$




**Grey $H_2$**

Steam methane reforming (50% $H_2$)
$CO_2$ is emitted into the atmosphere

SMR: $CH_4 + H_2O \xrightarrow{\Delta} CO + 3H_2$

**Black/brown $H_2$**

Gasification of coal (20% $H_2$)
$CO_2$ is emitted into the atmosphere

$3C + O_2 + H_2O \xrightarrow{\Delta} H_2 + 3CO$

**Blue $H_2$**

Steam methane reforming
$CO_2$ is captured & stored

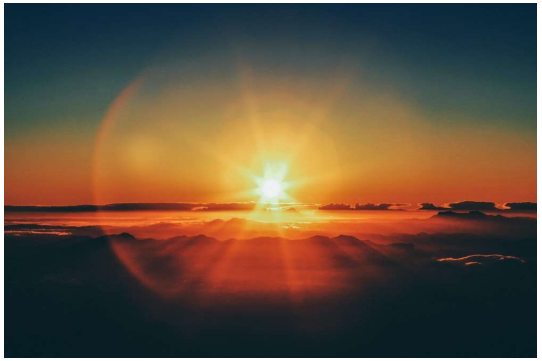
**Green $H_2$**

Water electrolysis (4% $H_2$)
Green electricity from solar/wind

$H_2O \xrightarrow{\text{Electricity}} H_2 + ½\ O_2$

# Solar Energy

Major power source
Solar power on Earth's surface is $10^{16}$ Watts



Light

Heat

Solar energy keeps the temperature of the earth's atmosphere at normal level
So that living things can survive
Heat causes currents in the atmosphere – temperature gradients lead to wind
Heat causes tidal waves in the Ocean
Causes the water cycle
Generates photosynthesis

Solar Energy Can be utilized in two ways

- Solar thermal…examples, solar water heater and solar cookers
- Photovoltaic

# Solar Thermal

Solar cooker

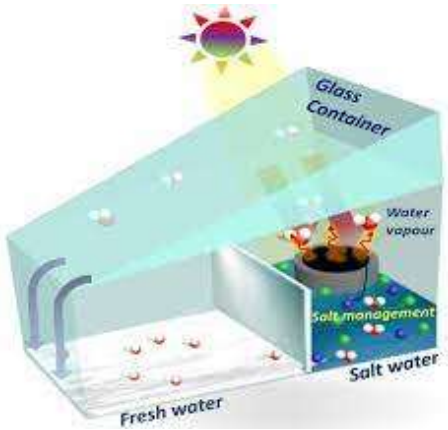





Solar engine for water pumping



Sea water desalination




Solar furnace



Solar streetlight







# Wind Energy

Wind Energy can be economically used for the generation of electrical energy



Winds are caused by two main factors

1. Heating and cooling of the atmosphere which generates convection currents.

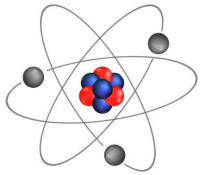
Heating is caused by the absorption of solar energy on the Earth's surface and in the atmosphere

2. The rotation of the earth with respect to atmosphere, and its motion around the sun

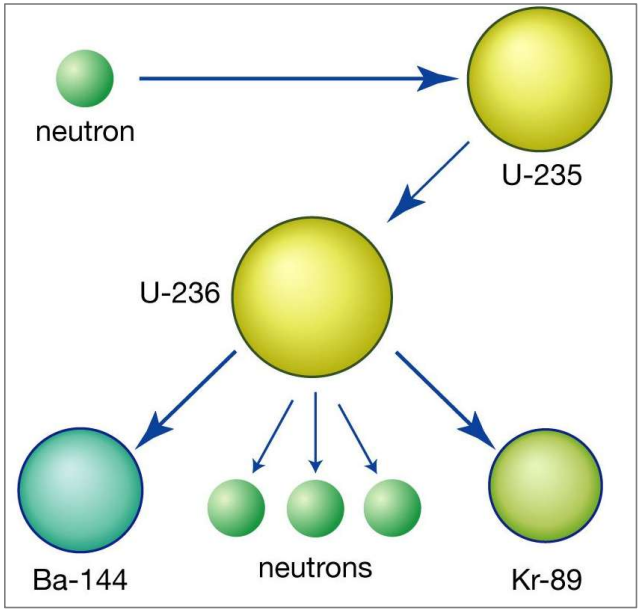
The energy available in the winds over the earth's surface is estimated to be $1.6 * 10^7$ MW

Wind energy drives a motor to produce electricity
Wind provides mechanical power, for pumping water

# Nuclear energy

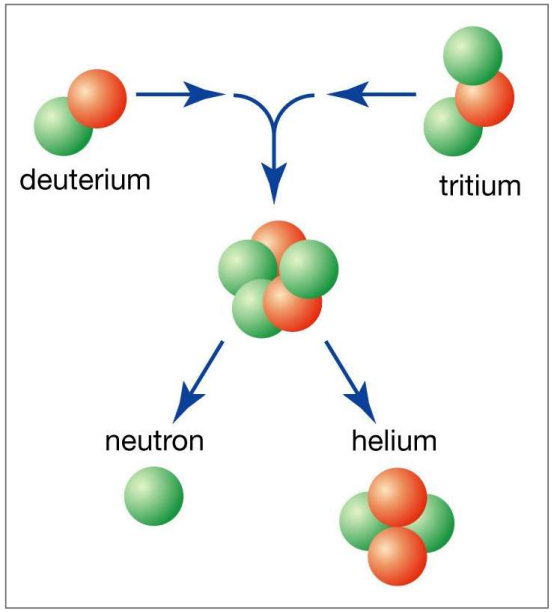


## Nuclear fission




$U^{235} + n^{1} \longrightarrow U^{236} \longrightarrow Ba^{144} + Kr^{89} + 3\ {}_{0}n^{1} + \text{Energy}$

## Nuclear fusion




${}_{1}H^{2} + {}_{1}H^{3} \longrightarrow {}_{2}He^{4} + {}_{0}n^{1} + \text{Energy}$

Four nuclear power stations in India: Tarapur (Maharashtra), Rana Pratap Sagar (Rajasthan), Kalpakkam (Tamilnadu), and Narora (U.P)

# Can nuclear energy be renewable?

- Seawater contains **4 billion metric tons of uranium**, nearly 1,000 times more uranium available in its ore **(Uraninite)**.

- The 4 billion tons of uranium in seawater would fuel **a thousand nuclear power plants** for a **100,000 years**.

- When uranium is extracted from seawater, the same amount is leached from the rocks (that contain uranium) via steady- state chemical reactions between the water and rocks.

- As a result, uranium extracted from seawater is replenished continuously, **so nuclear energy becomes as endless as solar, hydro, and wind, and it can be called as renewable energy.**

**ENERGY CONSERVATION** IS AN IMPORTANT OPTION IN PLANNING AN ENERGY POLICY FOR THE FUTURE

**Efficiency** of Energy use is going to make significant impact

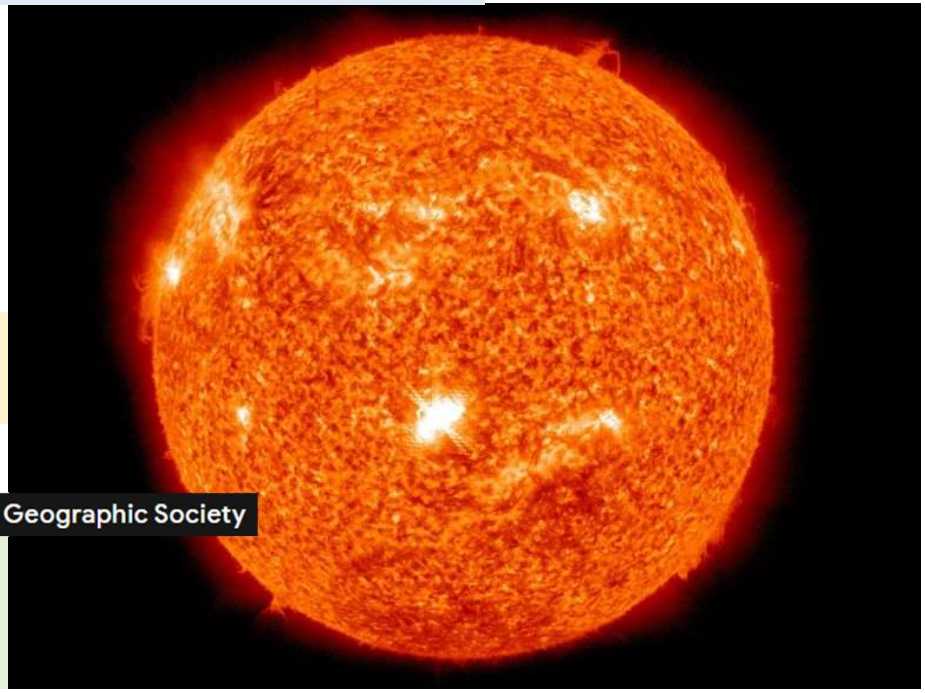
## Strategies to save Energy

- Greater use of mass transportation
- Less intensive use of energy for agriculture
- The manufacture of goods that are more durable,
- Greater use of energy-efficient appliances
- Recycling

# The Solar Radiation

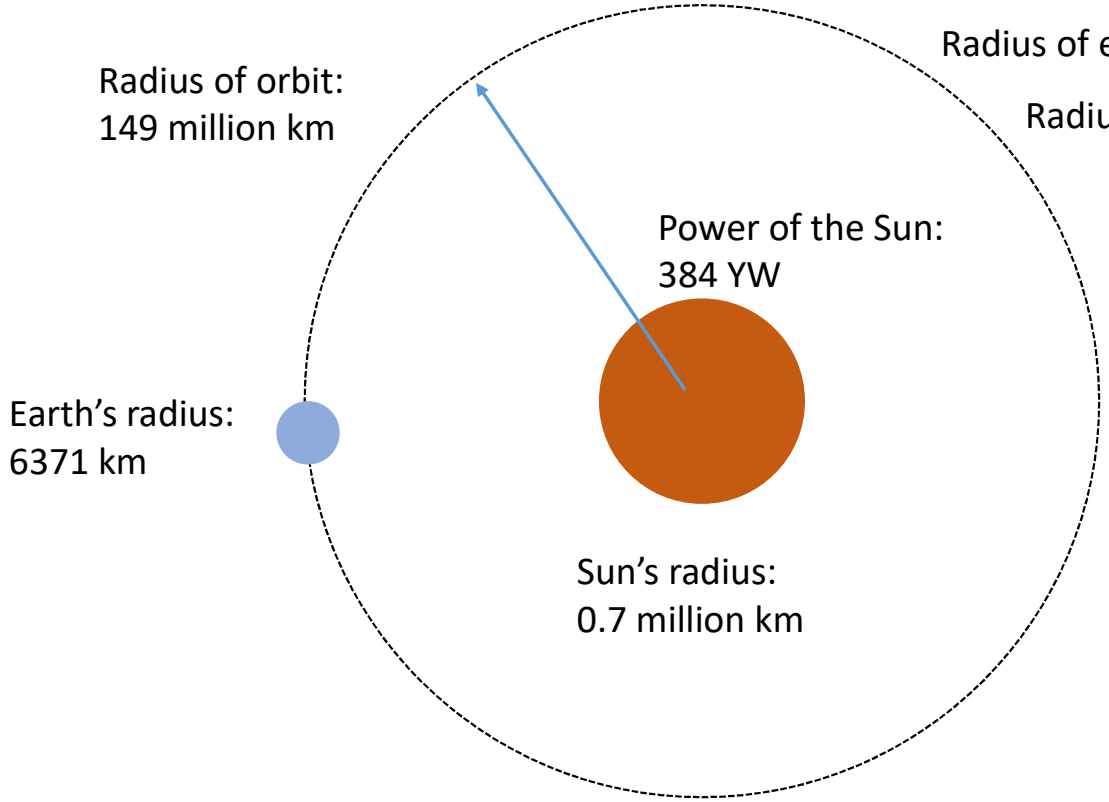
The most attractive source of energy is probably the sun. During daylight hours in sunny locations, huge quantities of solar energy reach the Earth's surface. This energy comes to us free and is **nonpolluting** and, for all practical purposes, **infinitely renewable**

Surface of the Sun ~ 5500 °C
Core of the Sun, several million °C



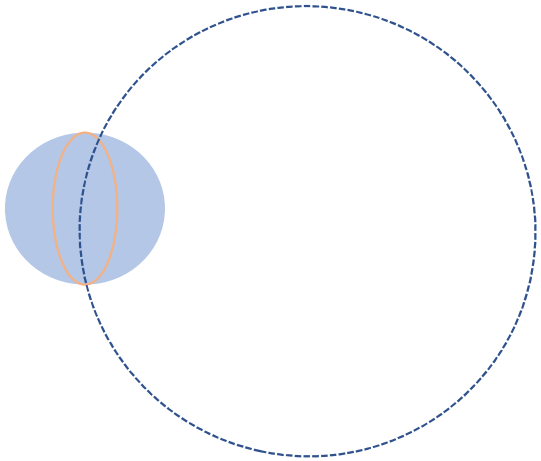

Sun gives out 384 Yotta Watt
= 384 x $10^{24}$ W
=3.84 x $10^{26}$ W

# Power of the Solar radiation




Intensity of Sun's radiation at Earth's orbit:
$= 3.84 \times 10^{26}/(4 \times 3.14 \times (1.49 \times 10^{11})^2 = 1377\ W/m^2$

# Power received by the Earth



Area of Earth's disc:
$= 3.14 \times (6.371 \times 10^6)^2 = 1.27 \times 10^{14}\ m^2$

Power received from the Sun, by Earth:
$1.27 \times 10^{14}\ m^2 * 1377 = 1.755 \times 10^{17}$ W or J/s

Energy received from the Sun, by Earth each year:
$=1.755 \times 10^{17}$ W x 60 x 60 x 24 x 365 = $5.5 \times 10^{24}$ J = 5.5 million Exa Joules per year

# The Scenario of Solar Energy

Global energy consumption :
= 500 Exa Joules per year

Earth receives from the Sun:
= 5.5 million Exa Joules per year

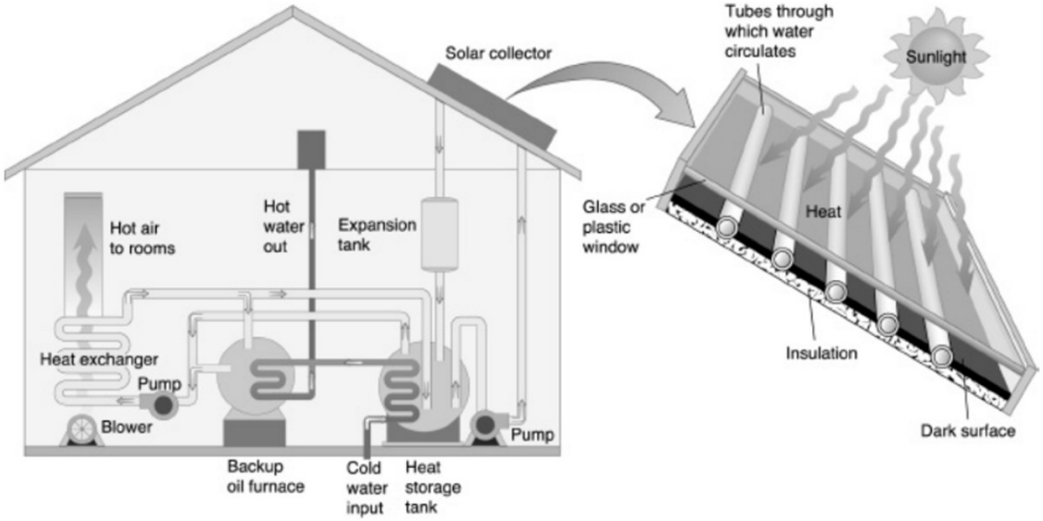
This is received in:
$500/5.5 \times 10^{6} = 9 \times 10^{-5}$ years = 0.033 days = 0.79 hours

Since 30% of the incident energy is reflected back, on the surface of the Earth, the energy used by humankind each year is received in:
= 0.79/0.70 ~ 1 hour

Earth receives nearly 5.5 million exa joules of energy from the Sun each year
The entire energy used by humankind each year, is received on the surface of the earth, from the sun, each hour

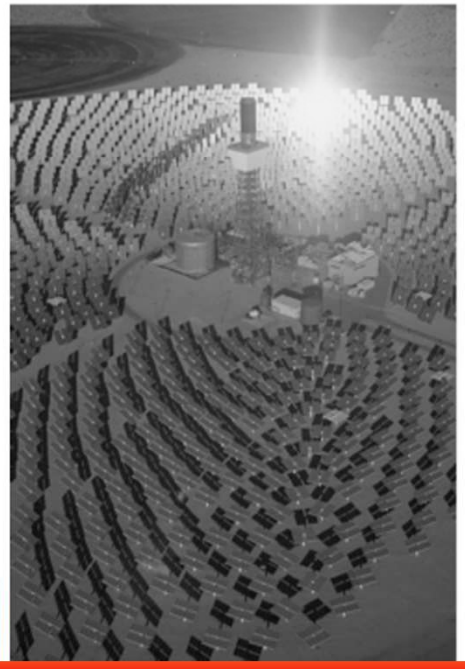
# STP plants generate electricity from solar thermal energy

## Active solar heating system



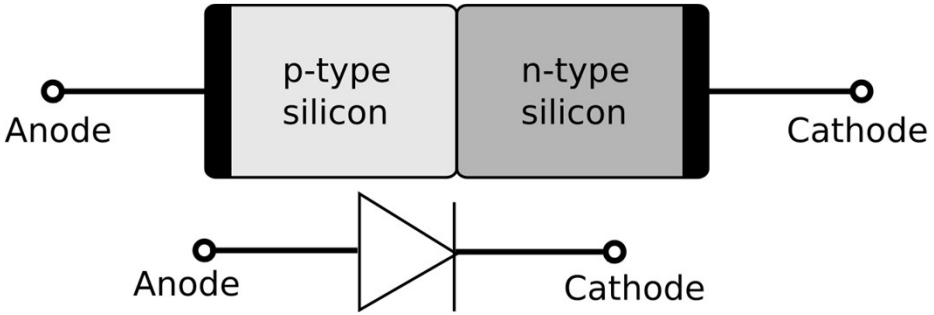


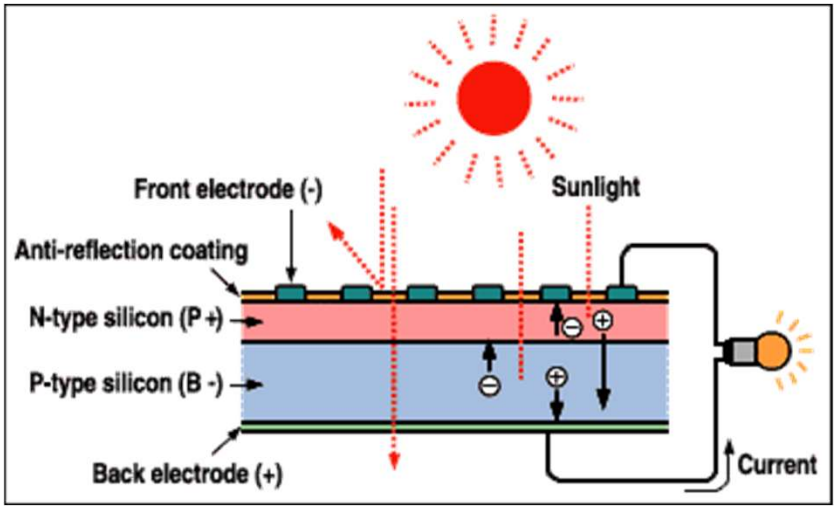






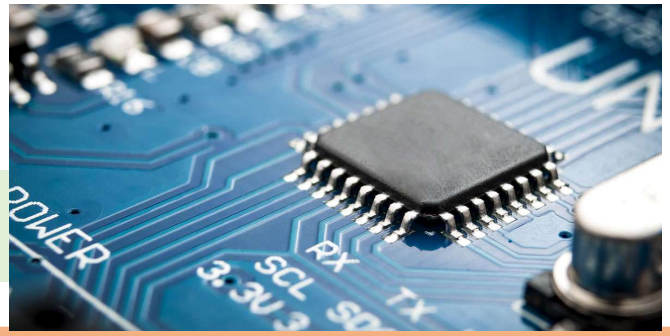
# Solar to electricity




Solar cell materials are chosen largely on the basis of how well their absorption characteristics match the solar spectrum and their cost of fabrication

Silicon fabrication technology is well developed as a result of its universality in the semiconductor electronics industry.



**The future of our planet will not be determined by how much energy we consume, but by how wisely we understand and use chemistry to produce and utilize that energy sustainably.**

**The environment is a complex chemical system in which air, water, land, and living organisms are interconnected. Understanding the chemistry of these systems enables us to use natural resources wisely, develop cleaner energy technologies, and make informed decisions for a sustainable future.**

- **Can modern society survive without fossil fuels? If yes, what would it take?**
- **If the Sun provides enough energy to meet humanity's annual demand in about one hour, why are we still dependent on coal and petroleum?**
- **Should chemistry be viewed as a contributor to environmental problems or as the key to solving them?**

# Take-Home Messages

1. **Earth supports life because of the interaction between the atmosphere, hydrosphere, lithosphere, and biosphere.**
   - These four systems continuously exchange matter and energy to maintain a habitable environment.
2. **Chemistry is fundamental to understanding environmental processes.**
   - From clean water and air quality to energy production and climate change, chemistry explains how our environment functions.
3. **Energy has driven human civilization, but our choice of energy sources determines environmental sustainability.**
   - Fossil fuels provide reliable energy but contribute significantly to pollution and greenhouse gas emissions.
4. **Renewable energy and hydrogen technologies are essential for a sustainable future.**
   - Solar, wind, nuclear, and green hydrogen can reduce carbon emissions and help meet growing global energy demands.
5. **Sustainability requires both scientific innovation and responsible human behaviour.**
   - Every individual can contribute through energy conservation, resource efficiency, recycling, and informed decision-making.

**1. Which of the following best explains why Earth can support life?**

A. It is the closest planet to the Sun.

B. It has abundant fossil fuels.

C. The atmosphere, hydrosphere, lithosphere, and biosphere interact to maintain suitable conditions for life.

D. It has the strongest gravitational field in the Solar System.

**2. Why is seawater unsuitable for direct human consumption?**

A. It contains too much dissolved oxygen.

B. It has a high concentration of dissolved salts.

C. It contains excessive nitrogen gas.

D. It has a low boiling point.

**3. Which energy source is considered renewable?**

A. Coal

B. Petroleum

C. Natural gas

D. Solar energy

**4. Which type of hydrogen has the smallest carbon footprint?**

A. Grey hydrogen

B. Black hydrogen

C. Green hydrogen

D. Blue hydrogen

**5. Which statement about fossil fuels is correct?**

A. They are renewable because they are continuously produced.

B. They are formed over millions of years from buried organic matter.

C. They produce only water upon combustion.

D. They do not contribute to greenhouse gas emissions.

**1. If the Sun provides enough energy to meet the world's annual energy demand in about one hour, why are fossil fuels still the dominant energy source?**

- Energy storage
- Intermittent nature of solar energy
- Existing infrastructure
- Economic considerations
- Technology limitations

**2. Why is hydrogen often called a clean fuel, even though most industrial hydrogen production currently releases carbon dioxide?**

- Water is the only emission during fuel cell operation.
- Environmental impact depends on the production method.
- Difference between grey, blue, and green hydrogen.

**3. Ocean water contains approximately 35 g of dissolved salts per litre, whereas drinking water contains less than 300 mg/L. Explain why high salinity makes seawater unsuitable for drinking.**

- Osmosis
- Kidney function
- Dehydration
- Human physiology

**4. Fossil fuels took millions of years to form, yet humans may consume most reserves within a few centuries. Why is this considered an unsustainable practice?**

- Rate of formation vs. rate of consumption
- Resource depletion
- Climate change
- Energy security

**5. Environmental problems such as climate change, water pollution, and acid rain are often described as "chemical problems." Explain why chemistry is essential for understanding and solving these issues.**

- Chemical reactions in the atmosphere
- Water chemistry
- Pollutant formation
- Catalysis
- Carbon capture
- Green chemistry